CEG5302 Group Project Report

EC Project Group 12

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1. ZDT3 with NSGA-II

ZDT3 is adopted to evaluate the performance of NSGA-II on problems with discontinuous Pareto fronts. The algorithm uses non-dominated sorting to assign ranks and crowding distance to maintain population diversity. Binary tournament selection is applied, and offspring are generated using simulated binary crossover (SBX, $\eta_c = 20$) and polynomial mutation ($\eta_m = 20$).

The experiment is conducted with population size $N = 100$, crossover probability $P_c = 0.9$, mutation probability $P_m = \frac{1}{30}$, and 500 generations.

As shown in Fig. 1, the obtained solutions clearly form multiple disconnected segments, which is consistent with the theoretical Pareto front of ZDT3. The solutions are well spread across all regions without noticeable clustering, indicating that the crowding distance mechanism effectively preserves diversity. In addition, all Pareto-optimal regions are successfully identified, demonstrating that non-dominated sorting is correctly implemented.

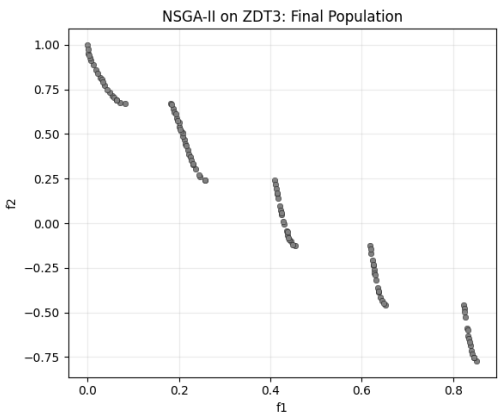

Figure 1: NSGA-II on ZDT3

Overall, the final population provides a good approximation to the true Pareto front in terms of both convergence and diversity.

Conclusion: NSGA-II is capable of handling discontinuous multi-objective problems and produces a well-distributed Pareto set on ZDT3.

2. Crashworthiness Design

In this task, NSGA-II was applied to a real-world crashworthiness design problem with three competing objectives: minimizing vehicle weight, acceleration characteristics, and toe-board intrusion. The decision variables correspond to the thickness of five structural components, each bounded between 1 and 3. Using the implemented NSGA-II algorithm, an initial population was randomly generated and evolved over multiple generations through non-dominated sorting, crowding-distance-based selection, simulated binary crossover, and polynomial mutation.

The algorithm was run with a population size of 100 and 200 generations. At the end of the optimization process, the final population consisted of 100 candidate solutions, each with 5 decision variables and 3 objective values. The obtained objective values were within reasonable

ranges (e.g., weight around 1660–1690, acceleration roughly between 6 and 11, and intrusion between 0.04 and 0.26), indicating that the implementation is functioning correctly. Moreover, the results clearly exhibit trade-offs among objectives. For example, some solutions achieve lower acceleration at the cost of higher intrusion, while others reduce intrusion but increase weight or acceleration. This confirms that NSGA-II successfully captures the conflicting nature of the objectives.

As shown in Figure 2, the final population forms an approximate Pareto front in the objective space. However, while the solutions are close to the Pareto front, their distribution is not perfectly uniform. Some regions appear denser than others, which is a known limitation of NSGA-II due to its reliance on crowding distance as a diversity metric. Overall, the results demonstrate that NSGA-II is effective in approximating the Pareto front for this engineering problem, providing a useful set of candidate designs for further decision-making.

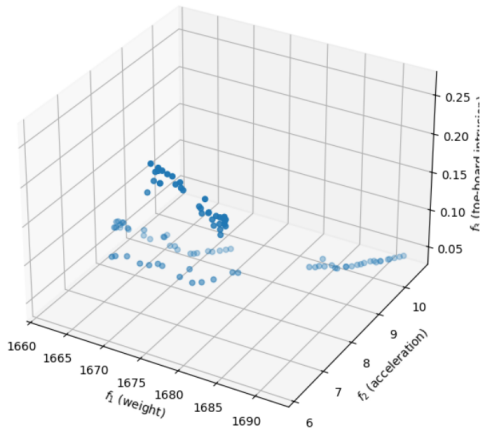

Figure 2: NSGA-II results on crashworthiness problem

3. MW7 with Constrained NSGA-II

MW7 is a constrained bi-objective benchmark with a disconnected feasible Pareto front. The constrained solver used here keeps the standard NSGA-II framework and modifies only the comparison rule so that feasibility is handled explicitly.

Let $c_j(x)$ denote the nonnegative violation of the jth constraint. The total constraint violation is

$$CV(x) = \sum_{j=1}^{m_c} c_j(x), \tag{1}$$

so feasible solutions satisfy $CV(x) = 0$. The constrained-dominance relation between two solutions x and y is

$$x \prec_c y \iff \begin{cases} CV(x) = 0,\ CV(y) > 0, & \text{or} \\ CV(x) = CV(y) = 0 \text{ and } x \prec y, & \text{or} \\ CV(x) > 0,\ CV(y) > 0 \\ \quad \text{and } CV(x) < CV(y), \end{cases} \tag{2}$$

where $x \prec y$ denotes the standard Pareto-dominance relation in objective space.

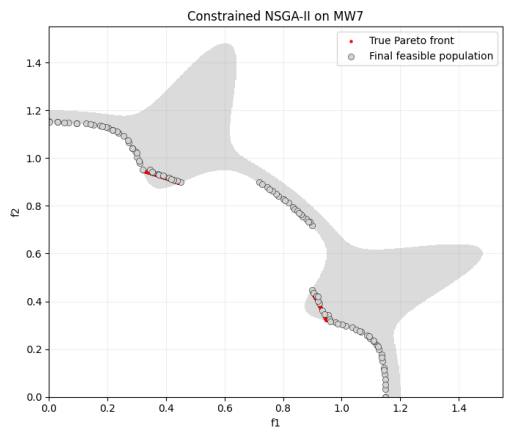

Figure 3: MW7 result in objective space.

The main run used a population size of 100, crossover probability of 0.9, mutation probability of 1/15, distribution indices of 20, and 500 generations. These settings were chosen as

a balanced baseline that keeps the runtime reasonable while still allowing the population to maintain feasible diversity.

Figure 3 shows the MW7 result in objective space. The feasible region, true Pareto front, and final population are shown together so that feasibility, convergence, and coverage can be judged at the same time. If some Pareto-front segments are missing, the most likely reason is the disconnected structure of the feasible front itself. Because the feasible pieces are separated by infeasible gaps, the population may converge to one region and lose representatives in others before they are fully explored. In addition, SBX crossover and polynomial mutation mainly act as local search operators, so rediscovering distant feasible segments can be difficult once they disappear.

4. RCM13: Gear Box Design

RCM13 is a constrained three-objective gear box design problem with seven decision variables, eleven nonlinear constraints, and a discrete variable $x_3 \in \{17, 28\}$. The constrained NSGA-II developed earlier was reused here, with only the problem definition and the repair step adapted to the gear box model.

Parameter analysis was carried out by varying one parameter at a time around the baseline setting $(N, p_c, p_m, G) = (120, 0.9, 1/7, 300)$. In all tested runs, the final population was fully feasible and the mean constraint violation was zero. The practical differences between settings therefore appeared mainly in the number of final non-dominated points and the runtime.

Table 1: Summary of the RCM13 parameter analysis.

Parameter	Tested values	Main observation	Selected
Population size N	60, 80, 100, 120, 160, 200	All runs remained fully feasible. Increasing N mainly increased the density of the final front, while runtime rose sharply from 52.39 s at $N = 60$ to 572.58 s at $N = 200$.	200
Crossover p_c	0.7, 0.8, 0.9, 1.0	Feasibility and front size were essentially unchanged across the tested range, which suggests low sensitivity to crossover probability on this problem.	0.9
Mutation p_m	0.05, 0.10, 1/7, 0.20, 0.30	All tested values produced fully feasible populations and very similar fronts. The standard setting $p_m = 1/7$ was retained as a balanced choice.	1/7
Generations G	100, 200, 300, 400, 500	Runtime increased almost linearly from 71.95 s to 347.56 s, while the final solution quality remained visually similar. This indicates that 300 generations already provide sufficient convergence.	300

Using the selection rule adopted in the code, the best setting was identified as $N = 200$, $p_c = 0.9$, $p_m = 1/7$, and $G = 300$. This choice maximized the number of feasible non-dominated solutions while preserving zero mean constraint violation. Figure 4 compares the selected setting with three representative alternatives: a smaller population $(60, 0.9, 1/7, 300)$, lower crossover probability $(120, 0.7, 1/7, 300)$, and higher mutation probability $(120, 0.9, 0.3, 300)$.

Figure 4 shows both the selected setting and three representative alternatives. All four

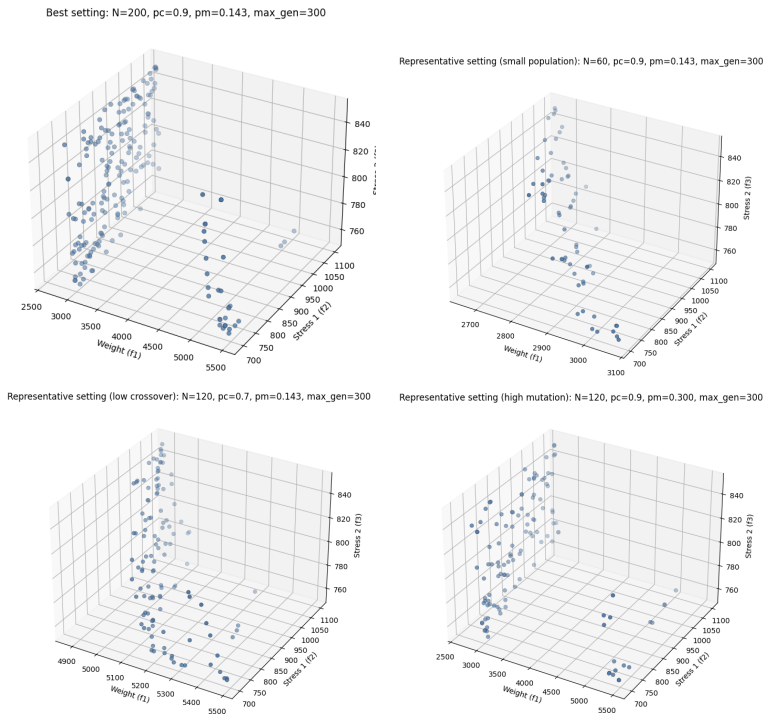

Figure 4: RCM13 final populations. Top-left: selected setting $(200, 0.9, 1/7, 300)$; top-right: small population $(60, 0.9, 1/7, 300)$; bottom-left: low crossover $(120, 0.7, 1/7, 300)$; bottom-right: high mutation $(120, 0.9, 0.3, 300)$.

runs produce a similar three-dimensional trade-off surface between weight and the two stress objectives, while the denser point cloud under $N = 200$ makes the final front visually more complete. This supports the conclusion that the constrained NSGA-II is stable on the gear box design problem and that population size is the most influential parameter in this set of experiments.

5. Conclusion

This project presents a unified workflow for multi-objective evolutionary optimization, from standard NSGA-II to constrained search. ZDT3 and crashworthiness design demonstrate the unconstrained framework, while MW7 and RCM13 highlight the importance of explicit feasibility handling. Overall, good results depend on convergence, diversity preservation, sensible parameter choices, and clear interpretation of the final trade-off set.